Adversarial Attack on Multimodal Trajectory Prediction

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Motivation: can 18 cm of driving fool a predictor?

Self-driving cars act on where they predict others will go, and an attacker controls one input: **its own path**. Following Zhang et al. [6], we attack our map-aware GNN predictor: how vulnerable is it, which part resists, and why?

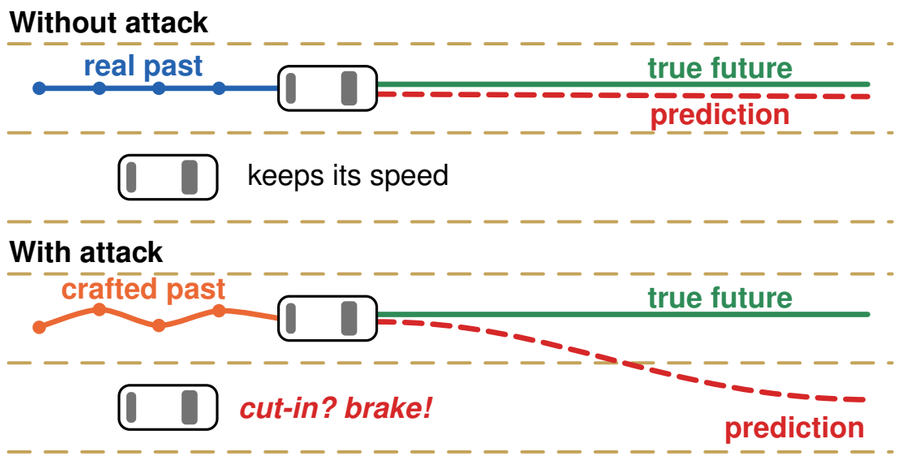


Figure 1: After [6]. The lower car predicts the upper car's path (red) and brakes for a cut-in that never comes.

Example: 15 cm per point forces a right turn

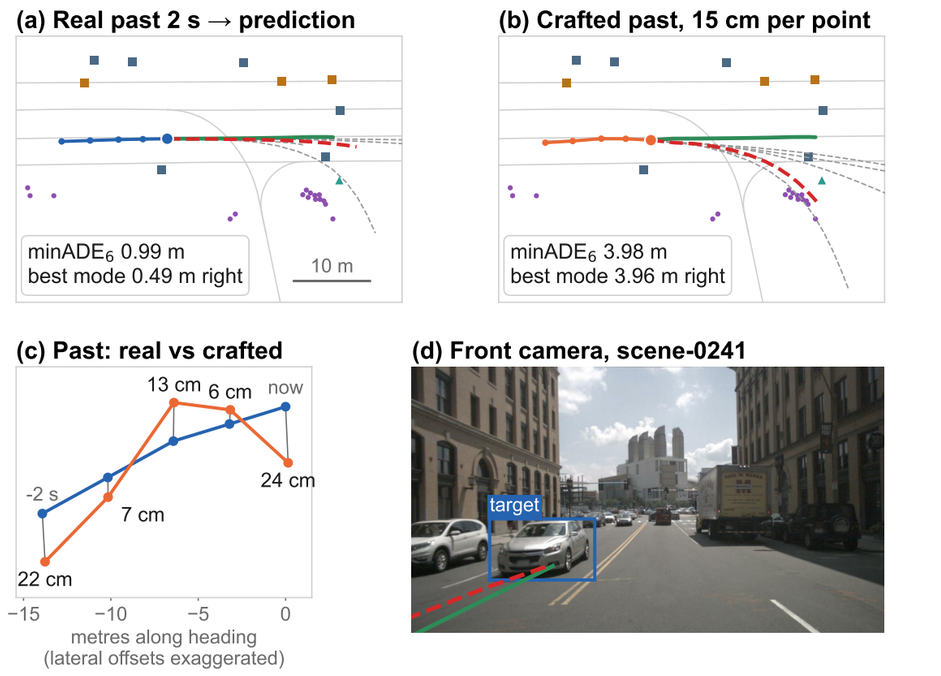


Figure 2: Final model (seed 42), nuScenes [2] scene-0241. Chosen for clarity: for this seed, the right attack crosses half a lane in 21.6% of samples.

Why is our model so easy to fool?

1. Short history, long horizon. Speed is read from 5 past points, so a shift grows over the 6 s forecast. Even constant velocity, which learns nothing, adds 21 m per metre of shift (ours: 7 m).

2. It trusts only what the attacker controls. Without the target's motion the error rises by 6.5 m, without neighbours by only ≈0.001 m: no other input can contradict a faked past.

3. Six modes move together. They share one context, so the whole fan shifts: under the ADE attack, samples with no mode within 2 m rise from 63% to 96%.



References

[1] S. Brody, U. Alon, and E. Yahav. How attentive are graph attention networks? In *ICLR*, 2022.

[2] H. Caesar, V. Bankiti, A. H. Lang, S. Vora, V. E. Liong, Q. Xu, A. Krishnan, Y. Pan, G. Baldan, and O. Beijbom. nuScenes: A multimodal dataset for autonomous driving. In *CVPR*, 2020.

[3] H. Cui, V. Radosavljevic, F.-C. Chou, T.-H. Lin, T. Nguyen, T.-K. Huang, J. Schneider, and N. Djuric. Multimodal trajectory predictions for autonomous driving using deep convolutional networks. In *ICRA*, 2019.

[4] J. Gao, C. Sun, H. Zhao, Y. Shen, D. Anguelov, C. Li, and C. Schmid. VectorNet: Encoding HD maps and agent dynamics from vectorized representation. In *CVPR*, 2020.

[5] A. Madry, A. Makelov, L. Schmidt, D. Tsipras, and A. Vladu. Towards deep learning models resistant to adversarial attacks. In *ICLR*, 2018.

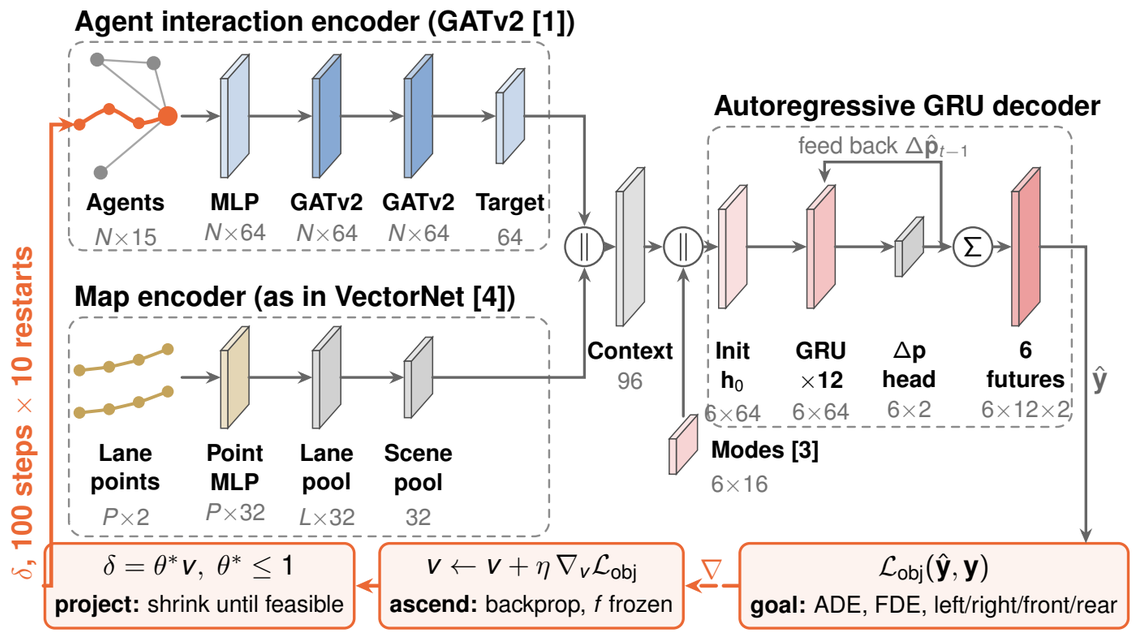
[6] Q. Zhang, S. Hu, J. Sun, Q. A. Chen, and Z. M. Mao. On adversarial robustness of trajectory prediction for autonomous vehicles. In *CVPR*, 2022.

Method: PGD attack through our predictor

$$\max_{\delta} \; \mathcal{L}_{\text{obj}}\big(f(\mathbf{x}+\delta,\ \text{neighbours},\ \text{map}),\ \mathbf{y}\big)$$

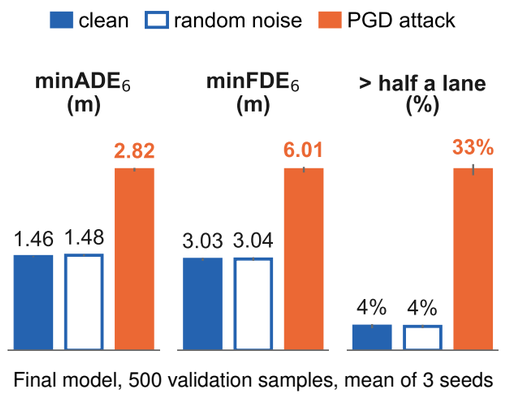
$$\text{s.t.} \;\; \|\delta_t\| \le 1\ \text{m per point}, \;\; \mathbf{x}+\delta \text{ physically feasible}$$

PGD [5] trains δ like a weight, with f frozen:



- **Feasible:** speed, acceleration and jerk within nuScenes bounds [6]; the crafted past must join the real future smoothly
- **Impact:** best of 6 pushed >1.85 m, half a lane [6]
- **Control:** smooth random noise of the same size

Results: is our final model vulnerable?



- **Yes:** PGD doubles both errors (minADE$_6$ ×1.93, minFDE$_6$ ×1.99) and 33% of forecasts cross half a lane (4% clean)
- **Direction, not noise:** same-size random noise (0.18 m per point) changes nothing
- Zhang et al. [6], averaged over their models and datasets: +167%/+150% ADE/FDE, 62% past half a lane; ours +93%/+99%, 33%

Results: which component adds robustness?

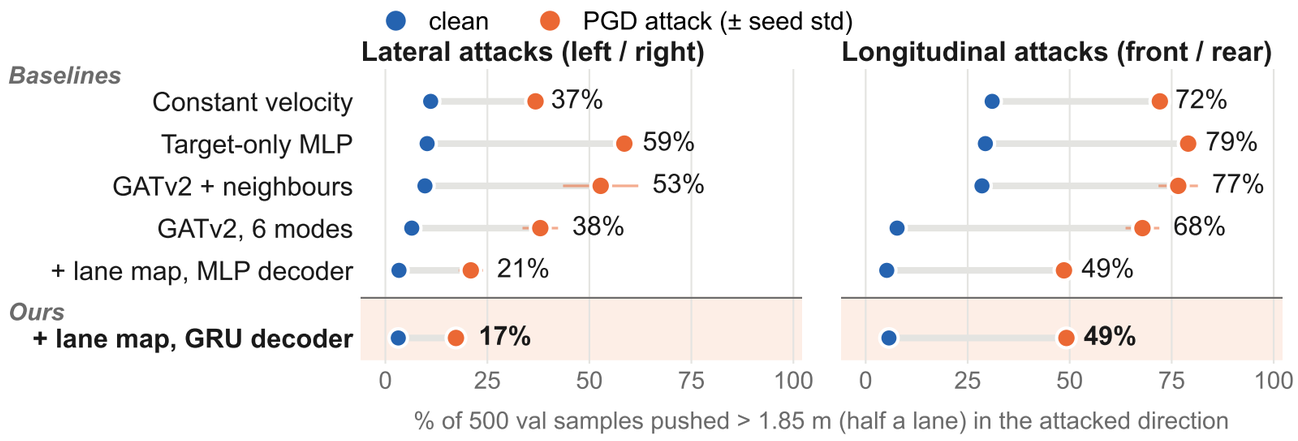


Figure 3: Same 500 validation samples and 3 seeds for every model. The first three predict one future; the rest use the best of 6.

- **The lane map helps most**, laterally: 38% → 21% past half a lane. Longitudinal attacks still reach 49%
- **Neighbours and decoder barely matter:** GATv2 + neighbours vs target-only MLP: 53% vs 59% laterally, 77% vs 79% longitudinally; most GATv2 runs ignore their neighbours. GRU vs MLP decoder: 17% vs 21% and 49% vs 49%

Conclusions

- 18 cm of crafted, physically feasible driving pushes a third of our final model's forecasts more than half a lane off.
- Context the attacker cannot move, the **lane map**, adds robustness.
- **Limitations:**
  - White-box, single frame, 500 validation samples; 1.85 m is only a proxy
  - 16% of real traces already break the physical bounds and mostly stay unperturbed
- **Next steps:**
  - Build a GNN predictor that genuinely learns from its neighbours
  - Test whether an attack on one road user spreads through the graph to others' predictions
  - Turn this into an evaluation framework for adversarial attacks on trajectory predictors

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